

5.1 Joint Probability Distributions (the continuous case)

A **joint probability density function** for the **continuous random variables X and Y** , denoted as $f_{XY}(x,y)$, satisfies the following properties:

- (1) $f_{XY}(x,y) \geq 0 \quad (\forall x, y)$
- (2) $\iint f_{XY}(x,y) dx dy = 1$
- (3) $P((X,Y) \in A) = \iint_A f_{XY}(x,y) dx dy$, for any region $A \subset \mathbb{R}^2$

Examples

7. (X,Y) is defined by $f_{XY}(x,y) = \frac{1}{8}$, for $0 \leq x \leq 2$, $0 \leq y \leq 4$. So (X,Y) is **uniformly** distributed over the rectangle with size 2 by 4.

A is the triangle with vertices $(0,0)$, $(2,0)$ and $(0,2)$. Determine $P((X,Y) \in A)$.

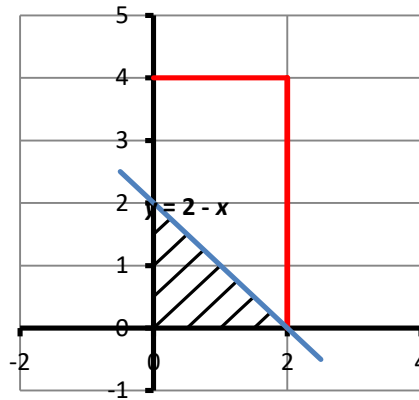


Figure 5.2 Region of integration for the probability that $(X,Y) \in A$

Solution

$$\begin{aligned}
 P(A) &= \iint_A f_{XY}(x,y) dx dy \\
 &= \int_0^2 \left\{ \int_0^{2-x} \frac{1}{8} dy \right\} dx \\
 &= \int_0^2 \frac{1}{8} \cdot (2-x) dx \\
 &= \left[\frac{1}{4}x - \frac{1}{16}x^2 \right]_0^2 \\
 &= \frac{1}{4}
 \end{aligned}$$

8. (X,Y) is defined by $f_{XY}(x,y) = cx$, for $0 \leq x \leq 3$, $0 \leq y \leq x$. Determine c . Then determine $P(X \leq 1)$.

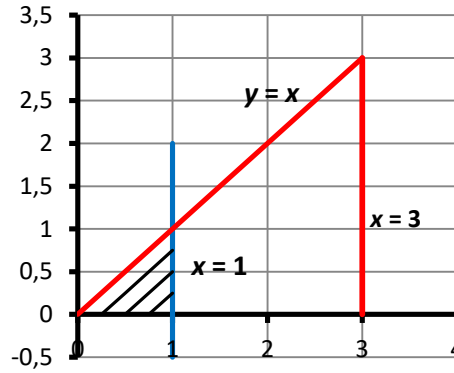


Figure 5.3 Region of integration for the probability that $X \leq 1$

Solution

$$\begin{aligned}
 \iint cx \, dx dy &= 1 & \Leftrightarrow & \int_0^3 \left\{ \int_0^x cx \, dy \right\} dx &= & 1 \\
 & & \Leftrightarrow & \int_0^3 cx^2 \, dx &= & 1 \\
 & & \Leftrightarrow & 9c &= & 1 \\
 & & \Leftrightarrow & c &= & \frac{1}{9}
 \end{aligned}$$

Thereafter

$$\begin{aligned}
 P(X \leq 1) &= \int_0^1 \left\{ \int_0^x \frac{1}{9}x \, dy \right\} dx \\
 &= \int_0^1 \frac{1}{9}x^2 \, dx = \frac{1}{27}
 \end{aligned}$$

Remark

Since $P(X \leq 1) = \int_0^1 f_X(x) \, dx$, the requested probability can also be calculated by first determining the **marginal probability density function $f_X(x)$** .

Definition

If the joint probability density function of X and Y is $f_{XY}(x, y)$, the **marginal probability density functions f_X and f_Y** are:

$$\begin{aligned}
 f_X(x) &= \int f(x, y) \, dy \\
 f_Y(y) &= \int f(x, y) \, dx
 \end{aligned}$$

Examples

9. See the data in Example 7. Determine $f_X(x)$ and $f_Y(y)$.

$$\begin{aligned}
 f_X(x) &= \int_0^4 \frac{1}{8} \, dy &= & \frac{1}{2} & , & \text{for } 0 \leq x \leq 2 \\
 f_Y(y) &= \int_0^2 \frac{1}{8} \, dx &= & \frac{1}{4} & , & \text{for } 0 \leq y \leq 4
 \end{aligned}$$

So $X \sim u[0; 2]$ and $Y \sim u[0; 4]$.

10. See the data in Example 8. Determine $f_X(x)$ and $f_Y(y)$.

$$\begin{aligned} f_X(x) &= \int_0^x \frac{1}{9}x \, dy = \frac{1}{9}x^2, & \text{for } 0 \leq x \leq 3 \\ f_Y(y) &= \int_y^3 \frac{1}{9}x \, dx = \left[\frac{1}{18}x^2 \right]_y^3 \\ &= \frac{1}{2} - \frac{1}{18}y^2, & \text{for } 0 \leq y \leq 3 \end{aligned}$$

$$\text{So } P(X \leq 1) = \int_0^1 \frac{1}{9}x^2 \, dx = \frac{1}{27}$$

Definition

The continuous random variables X and Y are independent, if:

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y), \quad \forall x, y$$

Examples (see the data in Examples 7, 8, 9 and 10)

$$11. \quad \left. \begin{aligned} f_{XY}(x, y) &= \frac{1}{8} \\ f_X(x) &= \frac{1}{2} \\ f_Y(y) &= \frac{1}{4} \end{aligned} \right\} \Rightarrow \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{4}$$

So X and Y are independent.

$$12. \quad \left. \begin{aligned} f_{XY}(x, y) &= \frac{1}{9}x \\ f_X(x) &= \frac{1}{9}x^2 \\ f_Y(y) &= \frac{1}{2} - \frac{1}{18}y^2 \end{aligned} \right\} \Rightarrow f_{XY}(3, 0) = \frac{1}{3} \neq f_X(3) \cdot f_Y(0) = 1 \cdot \frac{1}{2} = \frac{1}{2}$$

So X and Y are dependent.

5.2 Covariance and Correlation

$$\sigma_{XY} = \text{cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$$

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{V(X) \cdot V(Y)}}$$

Example 13

See the data in Example 9. Determine ρ_{XY} .

Solution

$$\begin{aligned} E(XY) &= \iint xy \cdot f(x, y) \, dx \, dy \\ &= \int_0^3 \left\{ \int_0^x xy \cdot \frac{1}{9}x \, dy \right\} dx \\ &= \int_0^3 \frac{1}{18}x^4 \, dx = \frac{27}{10} \end{aligned}$$

$$\begin{aligned} E(X) &= \int x \cdot f_X(x) dx \\ &= \int_0^3 x \cdot \frac{1}{9} x^2 dx = \frac{9}{4} \end{aligned}$$

$$E(Y) = \int_0^3 y \cdot \left(\frac{1}{2} - \frac{1}{18} y^2\right) dy = \frac{9}{8}$$

So

$$\text{cov}(X, Y) = \frac{27}{10} - \frac{9}{4} \cdot \frac{9}{8} = 0.16875$$

Verify that

$$E(X^2) = \frac{27}{5} \text{ and } E(Y^2) = \frac{9}{5}.$$

So

$$V(X) = \frac{27}{5} - \left(\frac{9}{4}\right)^2 = \frac{27}{80} = 0.3375 \text{ and } V(Y) = \frac{9}{5} - \left(\frac{9}{8}\right)^2 = \frac{171}{320} = 0.534375.$$

Thereafter

$$\rho_{XY} = \frac{0.16875}{\sqrt{0.3375 \cdot 0.534375}} = 0.3974$$

So there is **some positive linear relationship** between X and Y .

5.3 Bivariate Normal Distribution

The probability density function of a **bivariate normal distribution with parameters μ_X , σ_X , μ_Y , σ_Y and ρ** is:

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \cdot e^{-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x-\mu_X}{\sigma_X}\right)^2 - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 \right\}}$$

Remarks

- For the **marginal distributions** applies: $X \sim N(\mu_X; \sigma_X^2)$ and $Y \sim N(\mu_Y; \sigma_Y^2)$.
- $\rho = \rho_{XY}$. So the parameter ρ equals the **correlation coefficient** of X and Y .
- If $\rho = 0$ (i.e. X and Y are *uncorrelated*; so there is no *linear relationship*), then X and Y are **independent** (!).

Example 14

X and Y have a bivariate normal distribution with parameters $\mu_X = 5$, $\sigma_X = 1$, $\mu_Y = 3$, $\sigma_Y = 0.5$ and $\rho = 0$. Determine $P(X \geq 4, Y \geq 2)$.

Solution

$$\begin{aligned} P(X \geq 4, Y \geq 2) &= P(X \geq 4) \cdot P(Y \geq 2) \\ &= \text{normalcdf}(4, 10^{10}, 5, 1) \cdot \text{normalcdf}(2, 10^{10}, 3, .5) \\ &= 0.8222 \end{aligned}$$